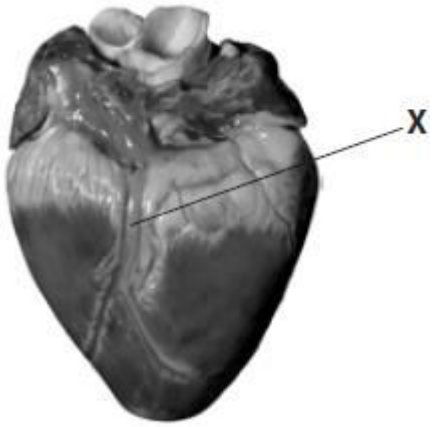
www.RocketRevise.com

Questions

Q1.

The photograph shows a mammalian heart.



RocketRevise

The blood vessel labelled **X** is a branch from the aorta.

The aorta is one of the major blood vessels of the heart.

(a) Explain why it is important that blood vessel **X** branches directly from the aorta.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

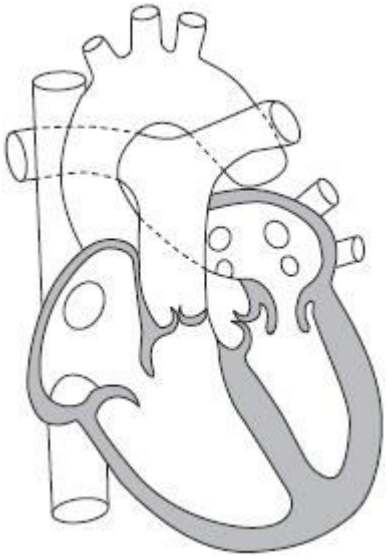
.....

RocketRevise

(b) The diagram shows the internal structure of a mammalian heart.

www.RocketRevise.com

www.RocketRevise.com



RocketRevise

(i) Explain which stage of the cardiac cycle is shown in this diagram.

(2)

.....

.....

.....

.....

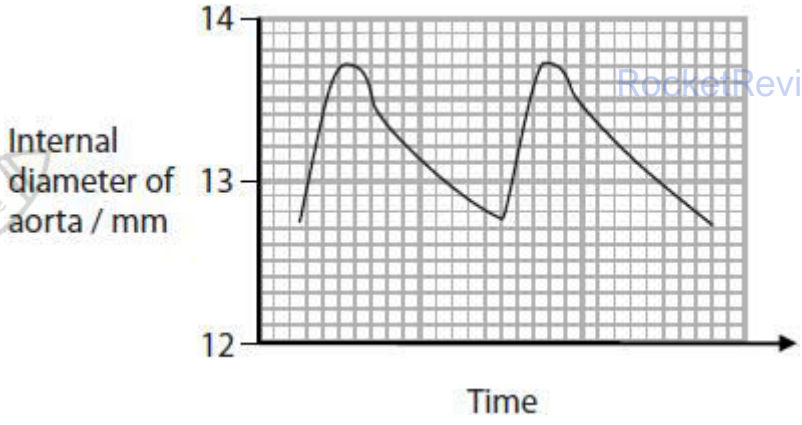
.....

.....

(ii) Draw arrows on the diagram to show the flow of blood through the left side of the heart and into the aorta.

(3)

(c) The graph shows the changes in the internal diameter of the aorta during two cardiac cycles.



RocketRevise

Explain the changes in the internal diameter of the aorta.

(3)

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

.....

(Total for question = 11 marks)

Q2.

Blood flow and blood pressure are affected by a number of factors.

Abnormal blood flow and abnormal blood pressure affect the health of a person.

(a) The graph shows the systolic pressure and the diastolic pressure as blood flows through human blood vessels.

RocketRevise

www.RocketRevise.com

www.RocketRevise.com



(i) Pulse pressure is the difference between systolic pressure and diastolic pressure.
In a healthy person, pulse pressure should be at least 25% of the systolic pressure.
Determine whether this person is healthy, using the point indicated by the arrow on the x-axis of the graph.

(2)

.....

.....

.....

.....

.....

.....

(ii) The mean arterial pressure (MAP) can be approximated using the formula

www.RocketRevise.com

www.RocketRevise.com

$$\text{MAP} = \text{diastolic blood pressure} + \frac{(\text{systolic blood pressure} - \text{diastolic blood pressure})}{3}$$

Estimate the MAP in the elastic arteries for this person.

(2)

RocketRevise

Answer ..... kPa

(iii) If the mean arterial pressure (MAP) value is less than 8 kPa, the force pushing the blood through the vessels will be too low.

Explain how a low MAP could affect a person.

(2)

.....

.....

.....

.....

.....

.....

(b) Blood flow depends on the change in pressure (ΔP), the radius of the blood vessel lumen (r), the length of the blood vessel (λ) and the viscosity of the blood (η).

Blood flow can be calculated using the formula

$$\text{Blood flow} = \frac{\pi \Delta P r^4}{8 \eta \lambda}$$

RocketRevise

Blood flow can also be calculated using the formula

$$\text{Blood flow} = \frac{\Delta P}{\text{resistance}}$$

(i) Which formula can be used to calculate resistance?

(1)

www.RocketRevise.com

www.RocketRevise.com

☐ **A** $\frac{8\eta\lambda}{\pi\Delta Pr^4}$

☐ **B** $\frac{\pi\Delta Pr^4}{8\eta\lambda}$

☐ **C** $\frac{\pi r^4}{8\eta\lambda}$

☐ **D** $\frac{8\eta\lambda}{\pi r^4}$

(ii) A change in which factor has the biggest effect on resistance?

(1)

RocketRevise

☐ **A** blood pressure
☐ **B** length of the blood vessel
☐ **C** radius of the blood vessel lumen
☐ **D** thickness of the blood vessel wall

(iii) Explain how arteries are adapted to reduce resistance to blood flow.

(2)

.....

.....

.....

.....

.....

.....

(c) Blood vessels can expand to accommodate increased blood flow. This is known as compliance.

(i) Explain how arteries are adapted to accommodate sudden increases in blood flow.

(2)

.....

.....

RocketRevise

.....

.....

.....

.....

(ii) Explain why a reduction in the compliance of a blood vessel can lead to cardiovascular disease (CVD).

(3)

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

(Total for question = 15 marks)

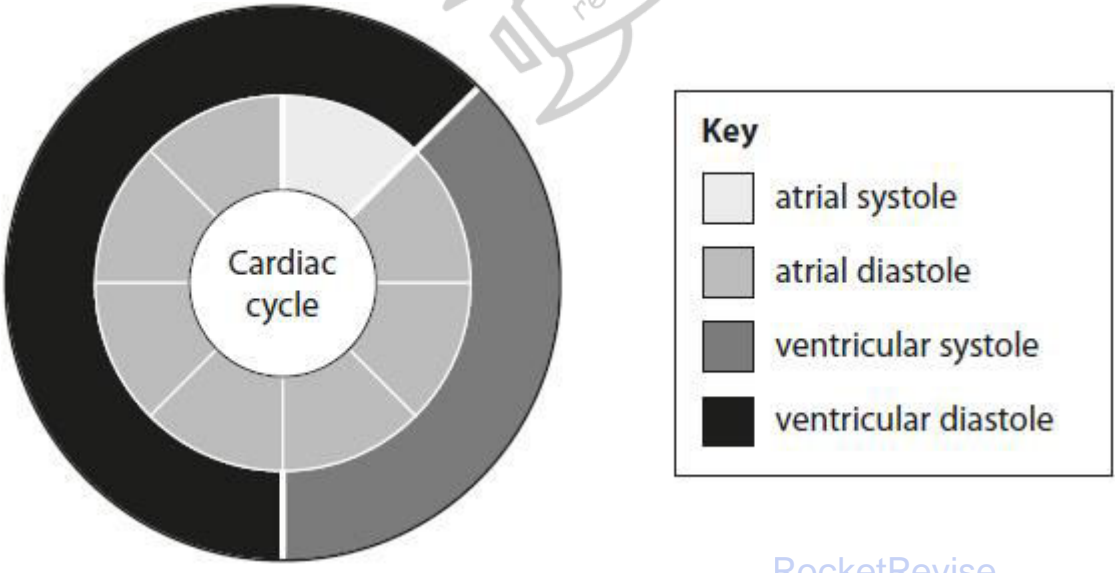
RocketRevise

Q3.

The cardiac cycle is the sequence of events that occurs when the heart beats.

A typical cardiac cycle takes 0.86 seconds.

(a) The diagram illustrates the cardiac cycle.



RocketRevise

(i) Which row of the table describes the atria and ventricles during atrial systole?

(1)

		Atria	Ventricles
☐	**A**	contracted	contracted
☐	**B**	contracted	relaxed
☐	**C**	relaxed	contracted
☐	**D**	relaxed	relaxed

(ii) Explain why there is a delay of 0.01 seconds between atrial systole and ventricular systole.

(2)

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

(iii) Using the information in the diagram, calculate the duration of ventricular systole in milliseconds. Express your answer in standard form.

(2)

RocketRevise

Answer ..... ms

(iv) State what proportion of the cardiac cycle is spent in ventricular diastole.

(1)

.....

.....

(b) A typical cardiac cycle takes 0.86 seconds.

During exercise, the heart rate increases and the duration of the cardiac cycle decreases. Calculate the increase in heart rate if the cardiac cycle decreases by 0.4 seconds.

(3)

RocketRevise

Answer ..... beats per minute

(Total for question = 9 marks)

www.RocketRevise.com

www.RocketRevise.com

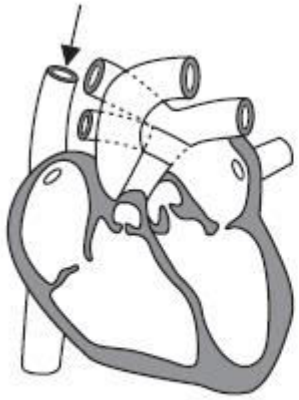
Q4.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

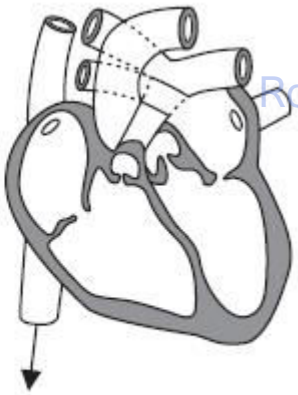
Blood flows through the heart during the cardiac cycle.

(a) Which diagram shows the direction of blood flow through the vena cava?

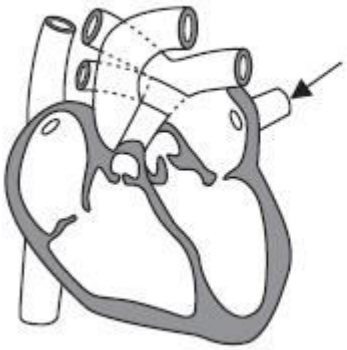
(1)



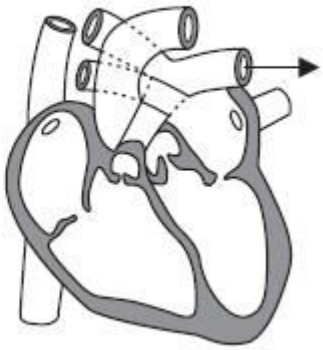
☐ **A**



☐ **C**



☐ **B**



☐ **D**

(b) Which row in the table describes the blood flow through the pulmonary artery and the pulmonary vein?

(1)

www.RocketRevise.com

www.RocketRevise.com

	Blood flow through pulmonary artery	Blood flow through pulmonary vein
☐ A	deoxygenated blood flowing away from the heart	oxygenated blood flowing towards the heart
☐ B	deoxygenated blood flowing towards the heart	oxygenated blood flowing away from the heart
☐ C	oxygenated blood flowing away from the heart	deoxygenated blood flowing towards the heart
☐ D	oxygenated blood flowing towards the heart	deoxygenated blood flowing away from the heart

RocketRevise

(c) The table describes the atrioventricular (AV) valves and the semilunar (SL) valves during the cardiac cycle.

Which description is true for each stage of the cardiac cycle?

(3)

Stage of the cardiac cycle	AV valves open and SL valves open	AV valves open and SL valves closed	AV valves closed and SL valves open	AV valves closed and SL valves closed
atrial systole	☐	☐	☐	☐
ventricular systole	☐	☐	☐	☐
diastole	☐	☐	☐	☐

(d) Describe how the events of the cardiac cycle change when the demand of body cells for oxygen increases.

(2)

.....

.....

.....

.....

.....

.....

RocketRevise

(Total for question = 7 marks)

Q5.

Many animals have a heart and circulatory system.

(a) Five litres of blood can pass through a human heart each minute.

www.RocketRevise.com

www.RocketRevise.com

Calculate the volume of blood that passes through this heart in 24 hours.
Give your answer in standard form.

(2)

Answer ..... litres

(b) Blood leaves the heart through the arteries.

RocketRevise

Compare and contrast the structure of the aorta with the structure of the pulmonary artery.

(2)

.....

.....

.....

.....

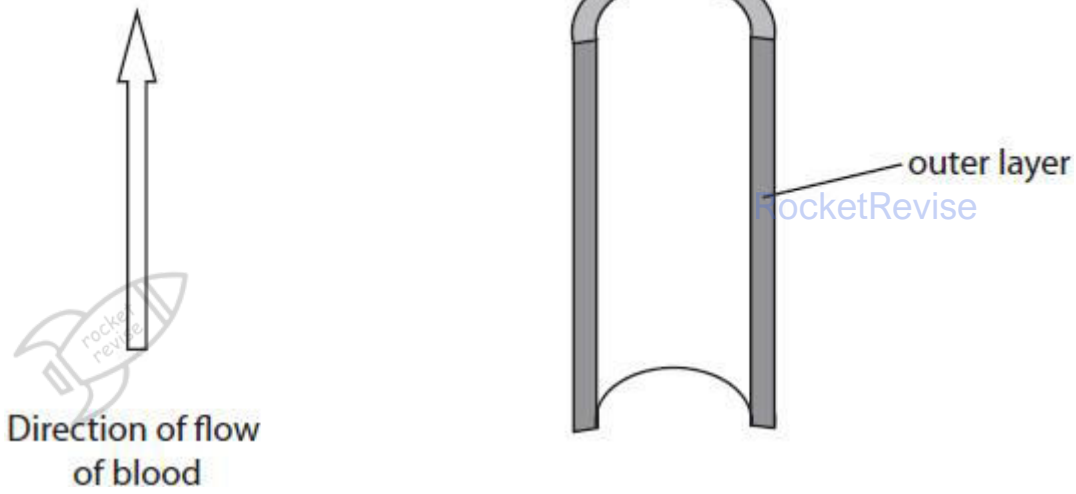
.....

.....

(c) Blood returns to the heart through the veins.

The diagram shows the outline of a vein. The direction of flow of blood is also shown.
Complete and label the diagram to show the structures present in a vein.

(3)



(Total for question = 7 marks)

www.RocketRevise.com

www.RocketRevise.com

Q6.

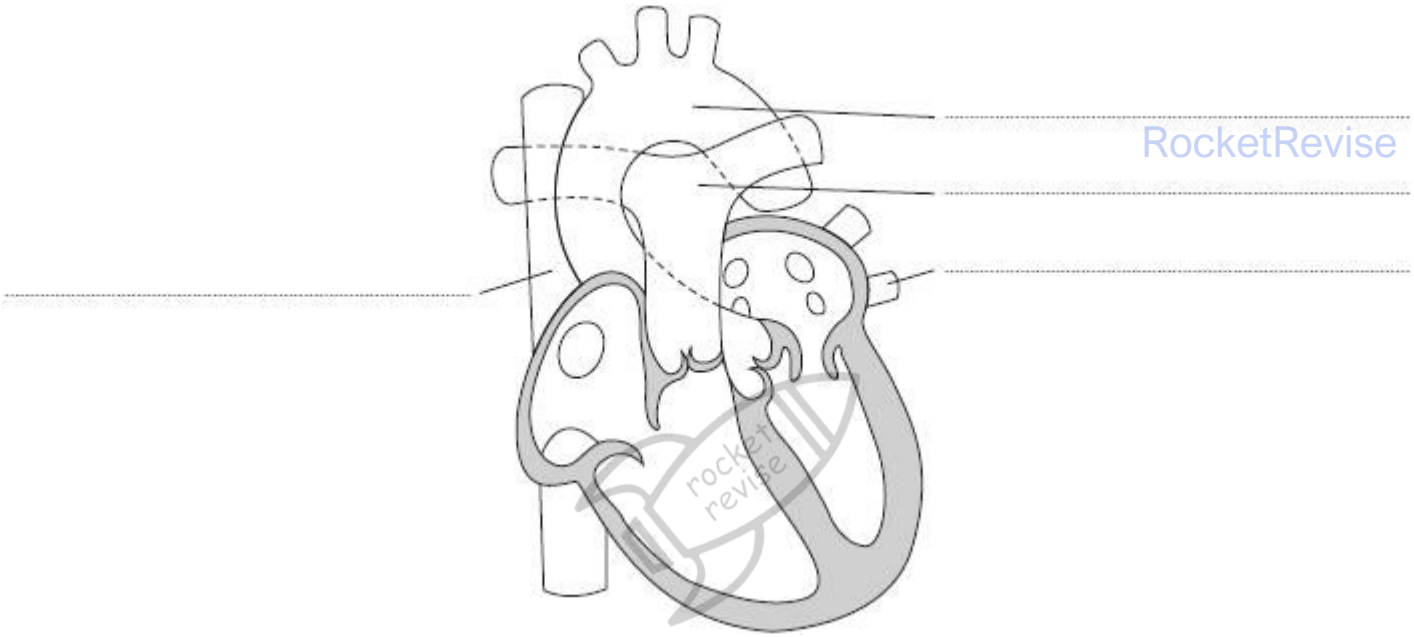
Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Many animals have a heart and circulation.

(a) The diagram shows the structure of a human heart.

Label the diagram with the names of the four major blood vessels.

(3)



(b) The table shows some structures and the types of blood vessel that they are found in.

Put a cross ☒ in each row to show where these structures are found.

(3)

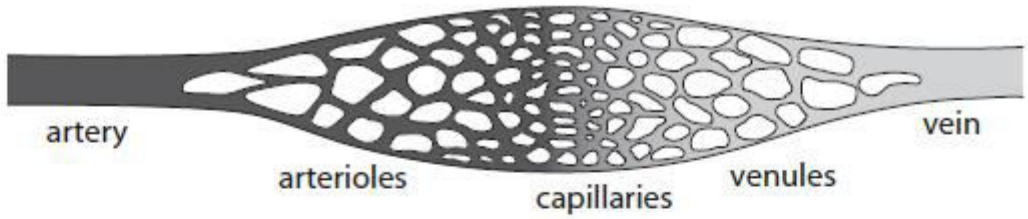
Structures	Found in arteries only	Found in capillaries only	Found in veins only	Found in arteries, capillaries and veins
Lining of endothelial cells	☐	☐	☐	☐
Valves along the length of the blood vessel	☐	☐	☐	☐
Wall only one cell thick	☐	☐	☐	☐

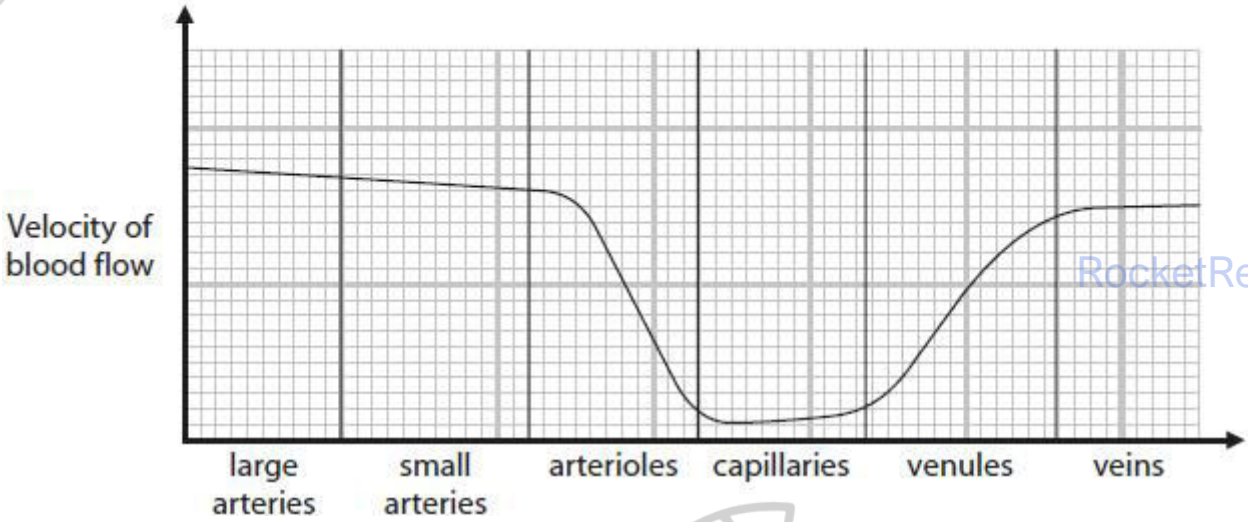
RocketRevise

(c) The graph shows the velocity of the blood as it flows through the arteries into the capillaries and then into the veins.

www.RocketRevise.com

www.RocketRevise.com





Describe the changes in the velocity of the blood as it flows from an artery to a vein.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

RocketRevise

(Total for question = 9 marks)

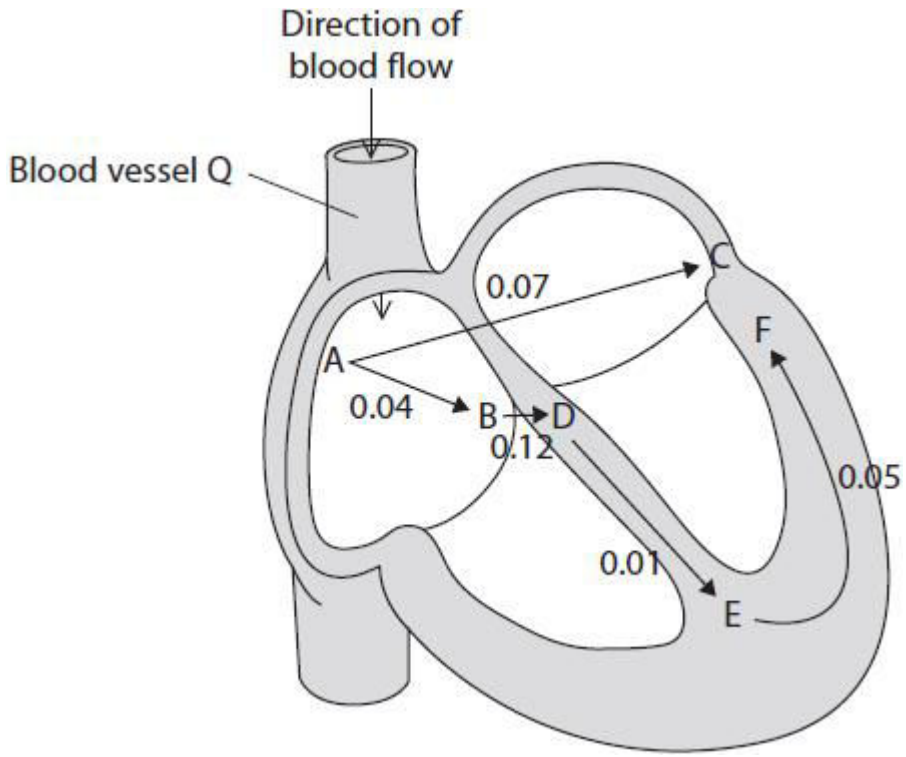
Q7.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The diagram shows the time taken, in seconds, for an impulse to travel through different parts of the heart.

www.RocketRevise.com

www.RocketRevise.com



RocketRevise

(a) What is the name of blood vessel Q?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(b) The cardiac cycle is divided into three stages. Two of these stages are atrial systole and ventricular systole.

(i) Name the other stage of the cardiac cycle.

(1)

.....

(ii) The duration of ventricular systole is the time it takes for the impulse to pass between E and F, shown on the diagram.

Calculate the proportion of time that the ventricles are contracting in one heartbeat if this heart beats 73 times per minute.

(2)

RocketRevise

Answer .....

(c) The part labelled B is in the wall of the atrium and the part labelled D is in the wall of the ventricles.

(i) The distance between B and D is 12.5 mm.

Calculate the speed at which the impulse travels between B and D.

(1)

www.RocketRevise.com

www.RocketRevise.com

Answer ..... mm s^{-1}

(ii) Explain why this speed needs to be slower than the speeds that the impulse travels between other parts of the heart.

(2)

.....

.....

.....

.....

.....

.....

RocketRevise

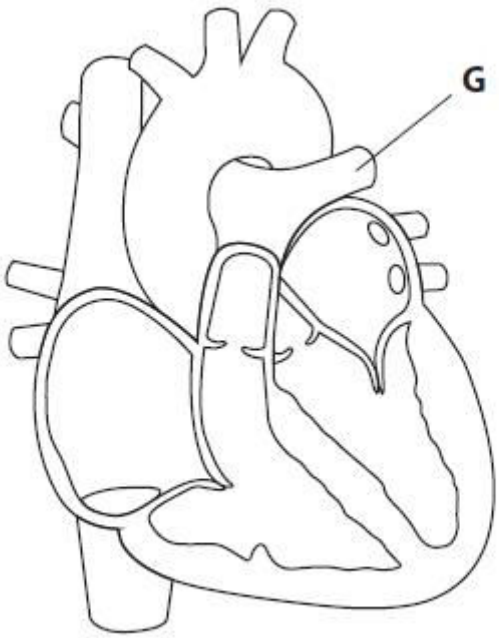
(Total for question = 7 marks)

Q8.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The cardiac cycle describes the events that take place in the heart during one complete heartbeat.

(a) The diagram shows a heart in one of the stages, stage **F**, of the cardiac cycle.



RocketRevise

(i) What is the name of the blood vessel labelled **G**?

(1)

- ☐ **A** aorta
- ☐ **B** pulmonary artery

www.RocketRevise.com

www.RocketRevise.com

☒ **C** pulmonary vein
☒ **D** vena cava

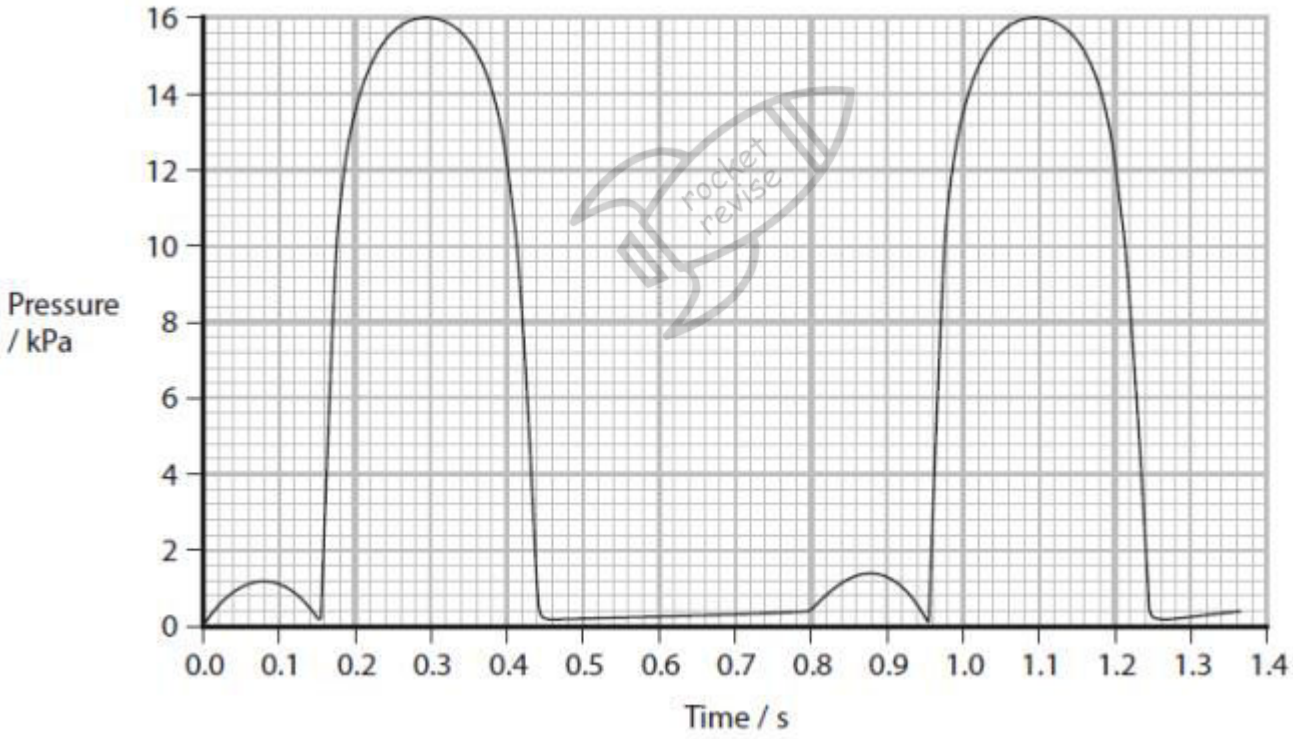
(ii) Which row of the table identifies the stages before and after stage **F**?

(1)

		Stage before stage F	Stage after stage F
☒	**A**	atrial systole	cardiac diastole
☒	**B**	atrial systole	ventricular systole
☒	**C**	ventricular systole	atrial systole
☒	**D**	ventricular systole	cardiac diastole

RocketRevise

(b) The graph shows pressure changes in the left ventricle of a person.



(i) Calculate the heart rate of this person.
Express your answer to 3 significant figures.

RocketRevise

(2)

Answer ..... beats min $^{-1}$

(ii) Another line could be drawn on this graph to show the pressure changes in the right ventricle.
Describe the shape and position of this line.
Give reasons for your answer.

(3)

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

(Total for question = 7 marks)

RocketRevise

Q9.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The structures of blood vessels relate to their functions.

The table gives information about some blood vessels in a dog.

Blood vessel	Number in body	Diameter / cm	Length of vessel / cm	Total surface area / cm^2	Velocity of blood flow / $cm\,s^{-1}$
aorta	1	1.0	40.0	1.3×10^2	28.0
large arteries	40	0.3	20.0	7.5×10^2	7.8
capillaries	1.2×10^9	8.0×10^{-4}	0.1	3.0×10^5	3.6×10^{-2}
large veins	40	0.6	20.0		1.9
vena cava	1	1.25		1.6×10^2	18.0

RocketRevise

(a) Complete the table.

Use the formula:
surface area = $2\pi rl$
where r is the radius of the blood vessel and l is the length of the vessel.

(3)

(b) Explain why the velocity of blood flow in the large arteries is slower than the velocity of blood flow in the aorta.

(3)

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

.....

.....

.....

RocketRevise

(c) Explain why the total surface area of the capillaries needs to be so high.

(2)

.....

.....

.....

.....

.....

.....

(d) (i) The diagrams show the relative proportions of components found in blood vessel walls.

Which diagram represents the wall of a large artery?

(1)

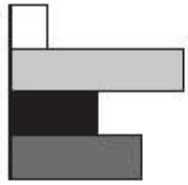
RocketRevise

www.RocketRevise.com

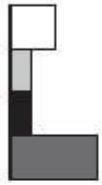
www.RocketRevise.com

(1)

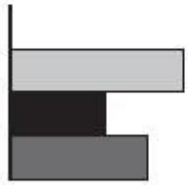
☒ A



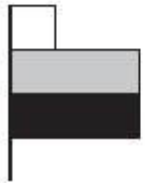
☒ B

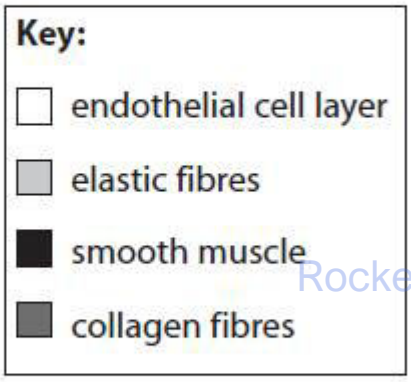


☒ C



☒ D

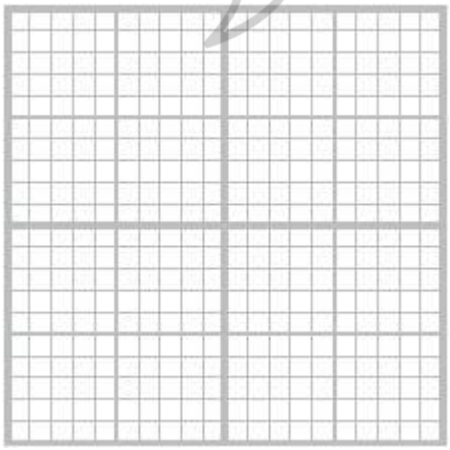




RocketRevise

(ii) Draw a diagram, using the style above, to represent the wall of a capillary.

(1)



RocketRevise

(Total for question = 10 marks)

Q10.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The blood clotting process has to be fast to prevent loss of blood from wounds.

(a) Which molecule traps red blood cells and platelets in the formation of a blood clot at a wound?

(1)

www.RocketRevise.com

www.RocketRevise.com

- **A** cholesterol
- **B** collagen
- **C** fibrin
- **D** fibrinogen

(b) How many of the following are converted into an active enzyme during blood clotting?

- fibrin
- fibrinogen
- prothrombin
- thrombin

RocketRevise

(1)

- **A** 1
- **B** 2
- **C** 3
- **D** 4

(c) How many of the following are soluble in blood plasma?

- fibrin
- fibrinogen
- prothrombin
- thrombin

(1)

- **A** 1
- **B** 2
- **C** 3
- **D** 4

(d) Thromboplastin is stored inside platelets.

Explain the advantages of storing thromboplastin inside platelets.

RocketRevise

(3)

.....

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

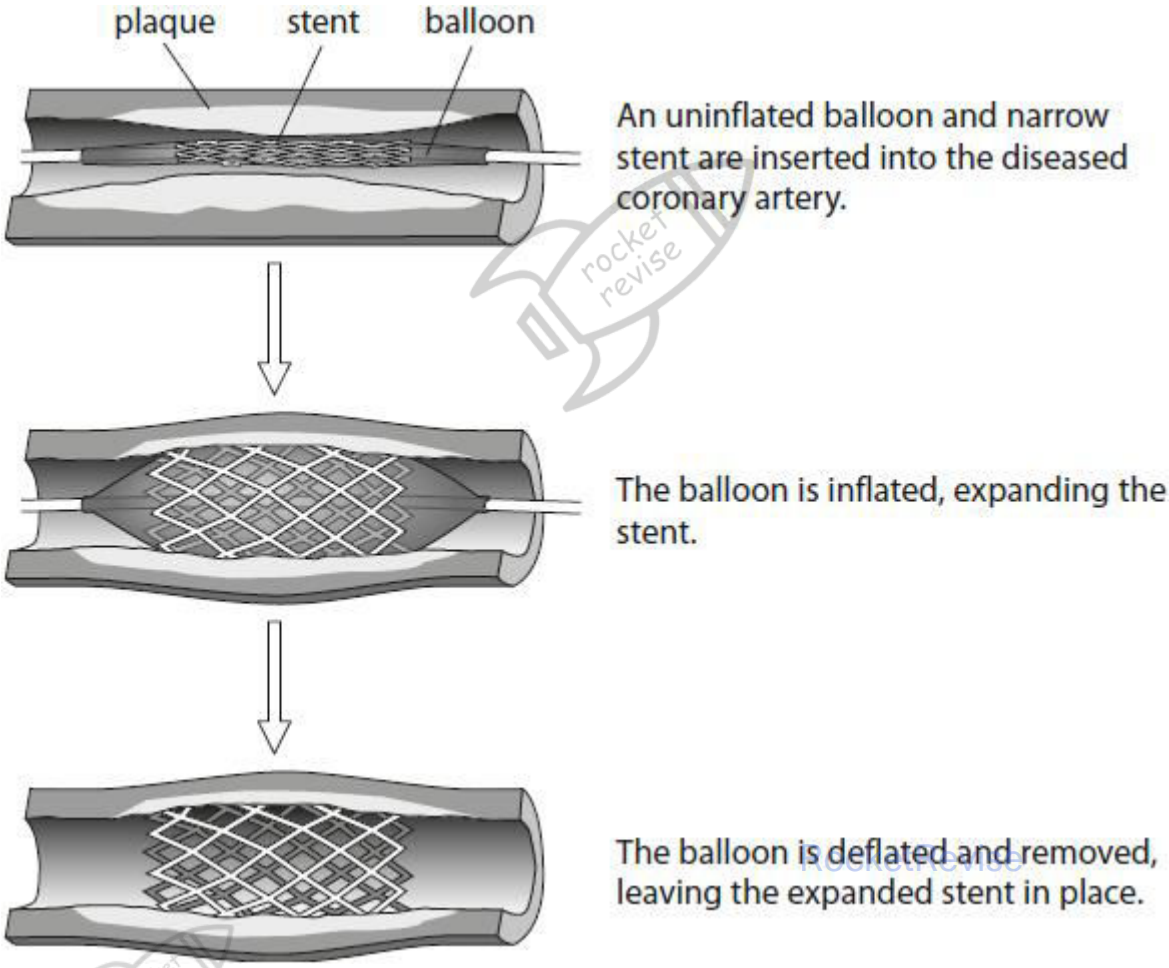
(Total for question = 6 marks)

Q11.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Stents are used in the treatment of atherosclerosis.

(a) The diagram shows how stents are positioned in a diseased coronary artery.



Explain why a stent is used in the treatment of atherosclerosis in a coronary artery.

(3)

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

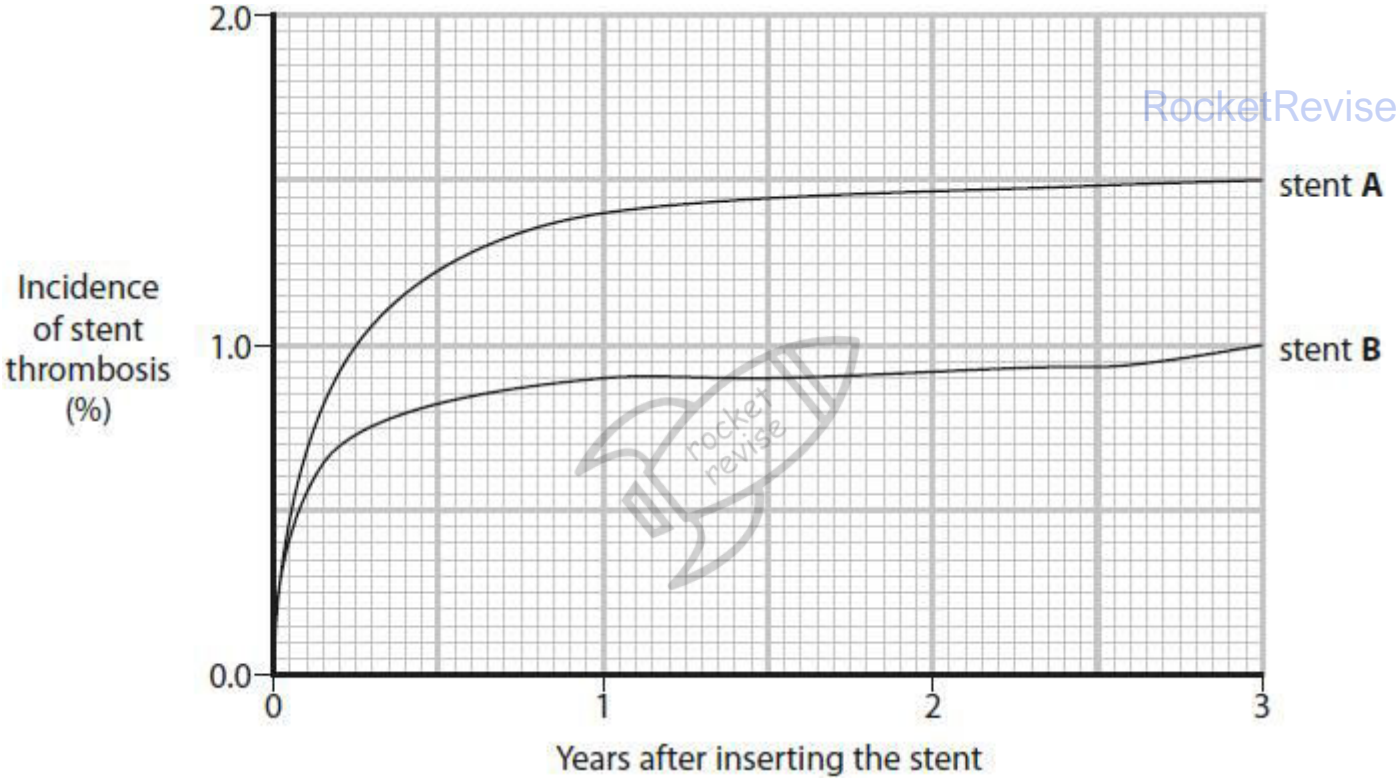
.....

.....

.....

(b) Inserting a stent can damage the artery.

This damage can result in stent thrombosis if the blood clotting process is stimulated.
A study looked at the damage caused by two different types of stent, stent **A** and stent **B**.
The graph shows the incidence of stent thrombosis found in this study.



(i) In this study, 800 patients had stent **A** inserted and 400 patients had stent **B** inserted.
Calculate the difference in the number of patients who developed stent thrombosis three years after inserting the stent.

(2)

Answer .....

(ii) State how the formation of a blood clot in the coronary artery could cause the death of the patient.

(1)

.....

.....

.....

(iii) Which components of the blood clotting process are enzymes in their active form?

(1)

- ☒ **A** prothrombin and thromboplastin only
- ☒ **B** prothrombin and thrombin only

www.RocketRevise.com

www.RocketRevise.com

- ☒ **C** prothrombin, thrombin and thromboplastin
- ☒ **D** thrombin and thromboplastin only

(iv) Which components of the blood clotting process are soluble in blood plasma?

(1)

- ☒ **A** fibrin and fibrinogen
- ☒ **B** fibrin and thromboplastin
- ☒ **C** fibrinogen and thromboplastin
- ☒ **D** fibrinogen only

(v) Stent **B** contains a drug to prevent stent thrombosis.
Suggest one type of drug that could be used in stent **B**.

(1)

.....

RocketRevise

(Total for question = 9 marks)

Q12.

The photograph shows a yak.



(Source: Zoonar GmbH / Alamy Stock Photo)

Yaks are cattle that live at high altitudes, such as in the Himalayas.

They are well-adapted for the low temperatures and low levels of oxygen in the air.

These adaptations include:

www.RocketRevise.com

www.RocketRevise.com

- a thick layer of fat below the skin
- large lungs and a large heart
- many small red blood cells
- presence of fetal haemoglobin throughout their adult life.

(a) Triglycerides are fats.

Describe the structure of a triglyceride.

(2)

.....

.....

.....

.....

.....

.....

(b) Explain why large lungs and a large heart would help a yak to survive at high altitudes.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

(c) The table shows some characteristics of the blood of yaks and the blood of humans.

Characteristic of blood	Blood of yaks	Blood of humans
Concentration of red blood cells / cells dm^{-3}	1.0×10^{13}	4.0×10^{6} to 6.0×10^{6}
Volume of one red blood cell / fl	43	90
Concentration of haemoglobin / g dm^{-3}	136	12 to 18

(i) The unit given for the volume of a red blood cell is a femtolitre (fl).

www.RocketRevise.com

www.RocketRevise.com

1 fl = 1.0×10^{-15} dm^3
Give the volume of a red blood cell of a yak in cm^3 .
Express your answer in standard form.

(1)

Answer ..... cm^3

(ii) The diagram gives some information about two spheres, sphere A and sphere B.

RocketRevise



- sphere A has a surface area : volume ratio of 1 : 1.3
- sphere B has a surface area of 768 mm^2

Calculate the surface area : volume ratio of sphere B.
Use the formula:

$$\text{Volume} = \frac{4}{3}\pi r^3$$

where $\pi = 3.0$

(2)

RocketRevise

Answer .....

* (iii) Discuss why the blood of yaks enables these animals to live at high altitudes.
Use all the information in this question to support your answer.

(6)

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for question = 14 marks)

Q13.

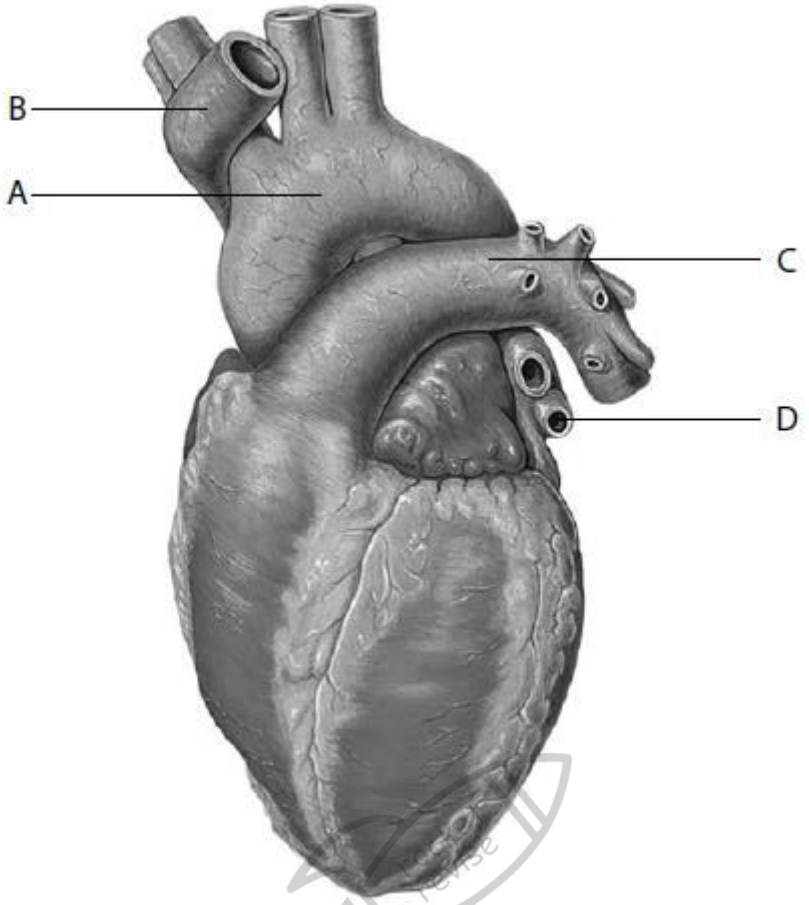
Mammals have a heart that pumps blood through a network of blood vessels.

(a) The drawing shows a human heart.

RocketRevise

www.RocketRevise.com

www.RocketRevise.com



RocketRevise

Used under Creative Commons Attribution-ShareAlike License - Patrick J Lynch

(i) Which blood vessel takes blood from the heart to the body?

(1)

- ☐ **A**
- ☐ **B**
- ☐ **C**
- ☐ **D**

(ii) Which blood vessel has semilunar valves and contains blood with the highest concentration of carbon dioxide?

(1)

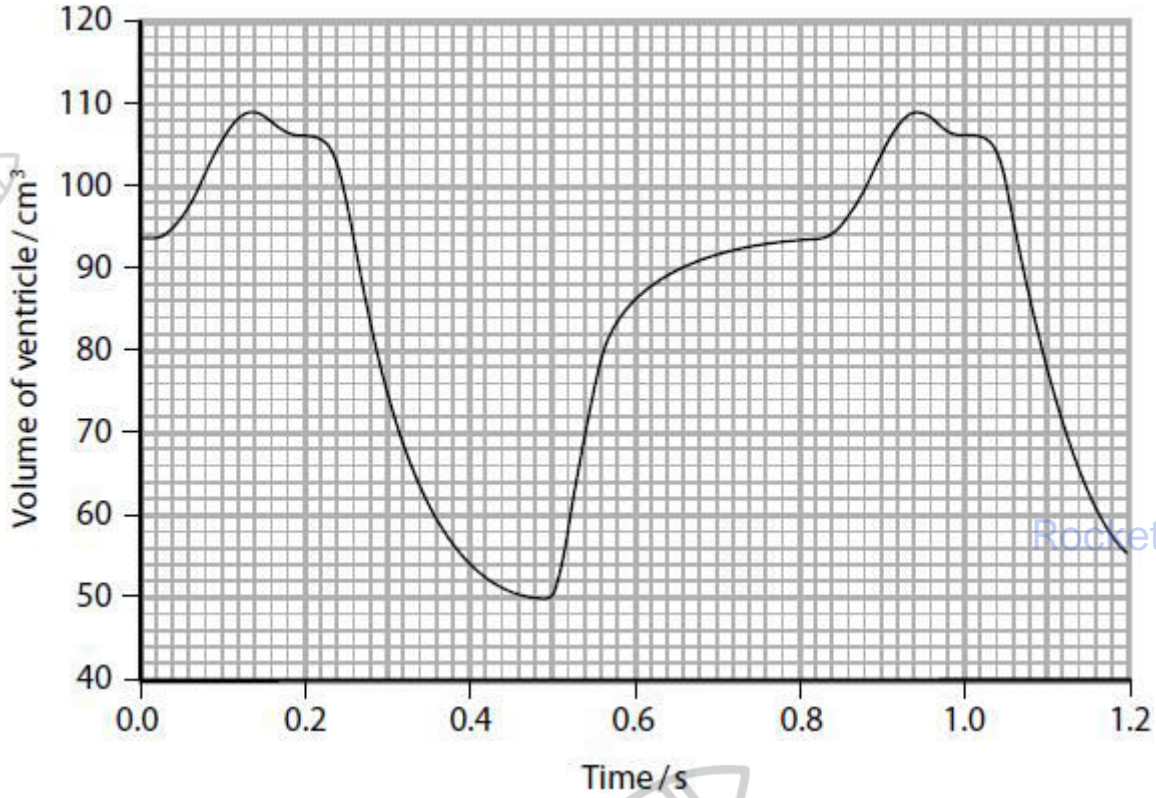
- ☐ **A**
- ☐ **B**
- ☐ **C**
- ☐ **D**

RocketRevise

(b) The graph shows the change in volume of the left ventricle during the cardiac cycle.

www.RocketRevise.com

www.RocketRevise.com



(i) When is this heart in ventricular systole?

(1)

- ☒ **A** at 0.1 seconds
- ☒ **B** at 0.4 seconds
- ☒ **C** at 0.6 seconds
- ☒ **D** at 0.8 seconds

(ii) Calculate the volume of blood in dm^3 that will be pumped out of this heart by the left ventricle each minute.

(3)

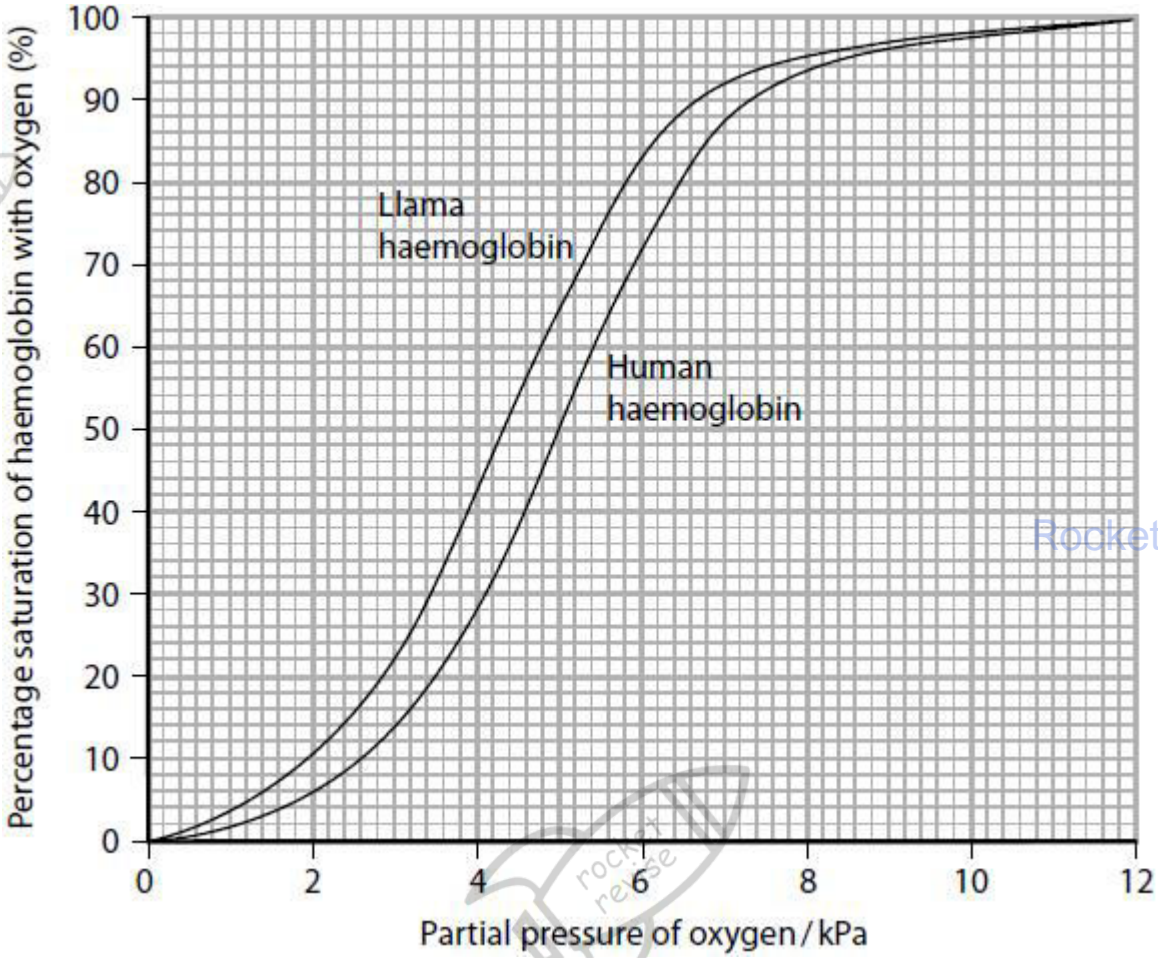
Answer ..... dm^3

(c) As altitude (height above sea level) increases, the partial pressure of oxygen in the air decreases.

Llamas are mammals that are adapted to living at high altitudes.
The graph shows the oxygen dissociation curves for llama haemoglobin and human haemoglobin.

www.RocketRevise.com

www.RocketRevise.com



Explain the differences between the dissociation curves. Use the information in the graph to support your answer.

(4)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for question = 10 marks)

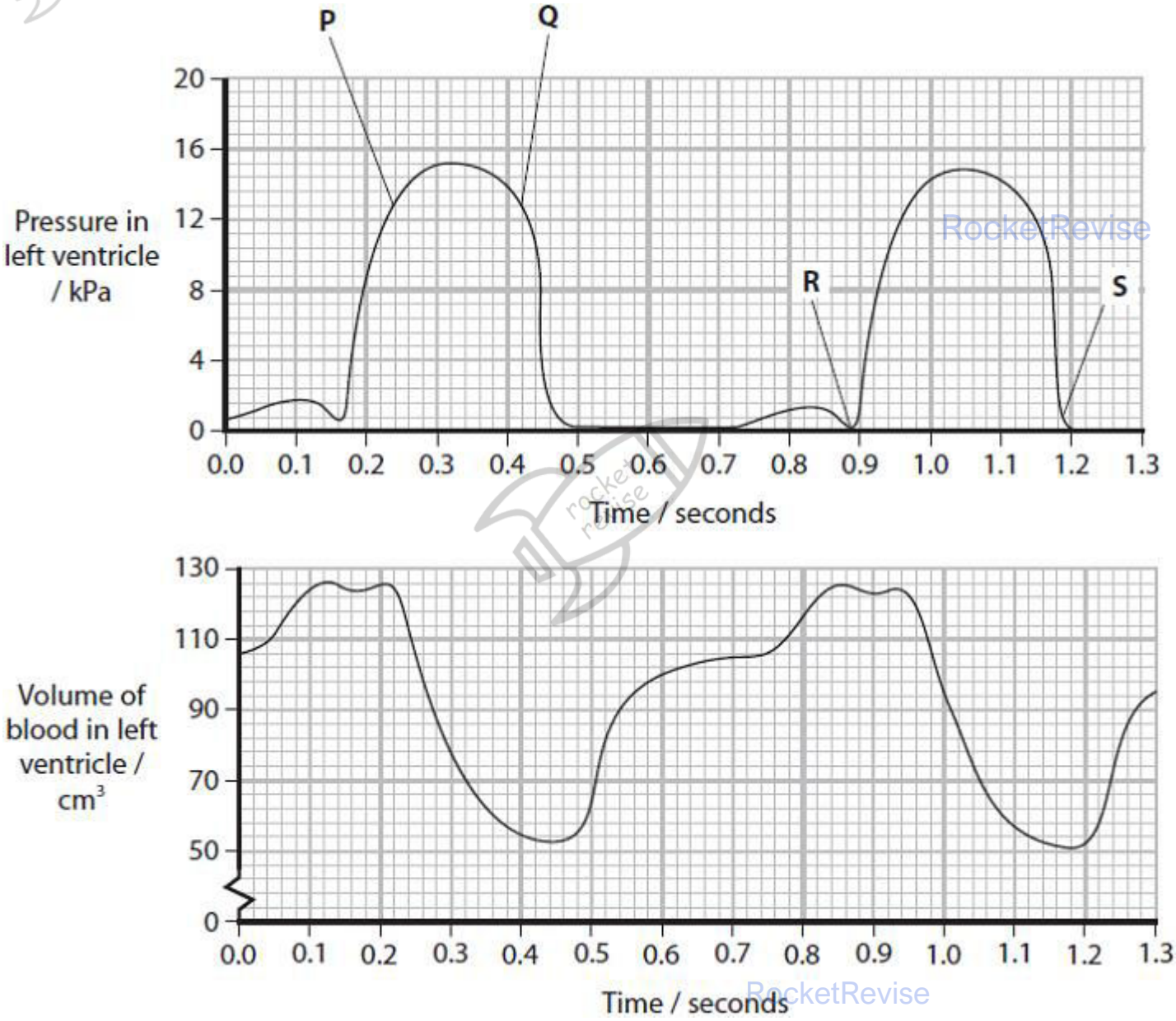
Q14.

www.RocketRevise.com

www.RocketRevise.com

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The graphs show how the pressure and volume of blood in the left ventricle change during the cardiac cycle.



(a) Which letter indicates when the atrioventricular (mitral) valves close?

(1)

- ☒ **A** P
- ☒ **B** Q
- ☒ **C** R
- ☒ **D** S

(b) Calculate the time for one cardiac cycle.

(1)

www.RocketRevise.com

www.RocketRevise.com

Answer ..... seconds

(c) The volume of blood in the left ventricle changes during the cardiac cycle.
Explain these changes.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

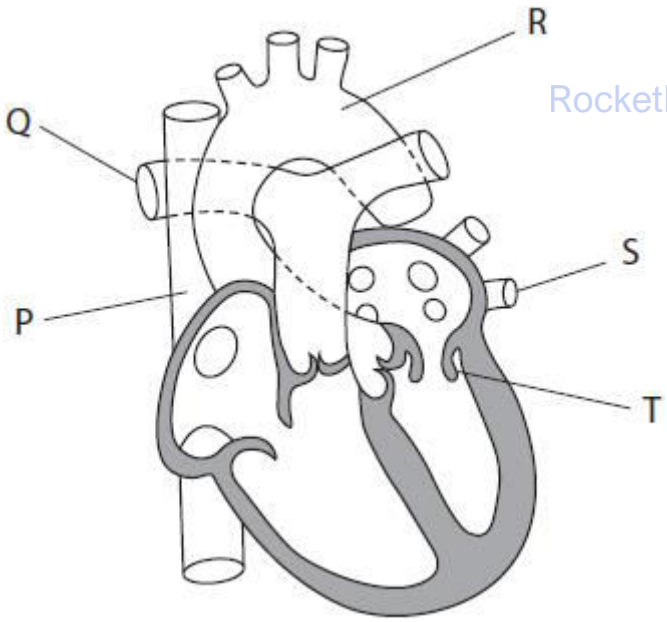
.....

(Total for question = 5 marks)

Q15.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

(a) The diagram shows the internal structure of a mammalian heart.



www.RocketRevise.com

www.RocketRevise.com

(i) How many of the following are correct for blood vessel P?

(1)

- carries blood to the heart
- has valves along its length
- has an endothelial lining

☒ **A** 0
☒ **B** 1
☒ **C** 2
☒ **D** 3

(ii) Why is blood vessel Q classified as an artery?

RocketRevise

(1)

☒ **A** because it carries deoxygenated blood
☒ **B** because it carries oxygenated blood
☒ **C** because it carries blood away from the heart
☒ **D** because it carries blood back to the heart

(iii) Which blood vessel does the coronary artery branch from?

(1)

☒ **A** P
☒ **B** Q
☒ **C** R
☒ **D** S

(iv) When in the cardiac cycle is valve T open?

(1)

☒ **A** atrial systole and cardiac diastole
☒ **B** atrial systole only
☒ **C** cardiac diastole and ventricular systole
☒ **D** ventricular systole only

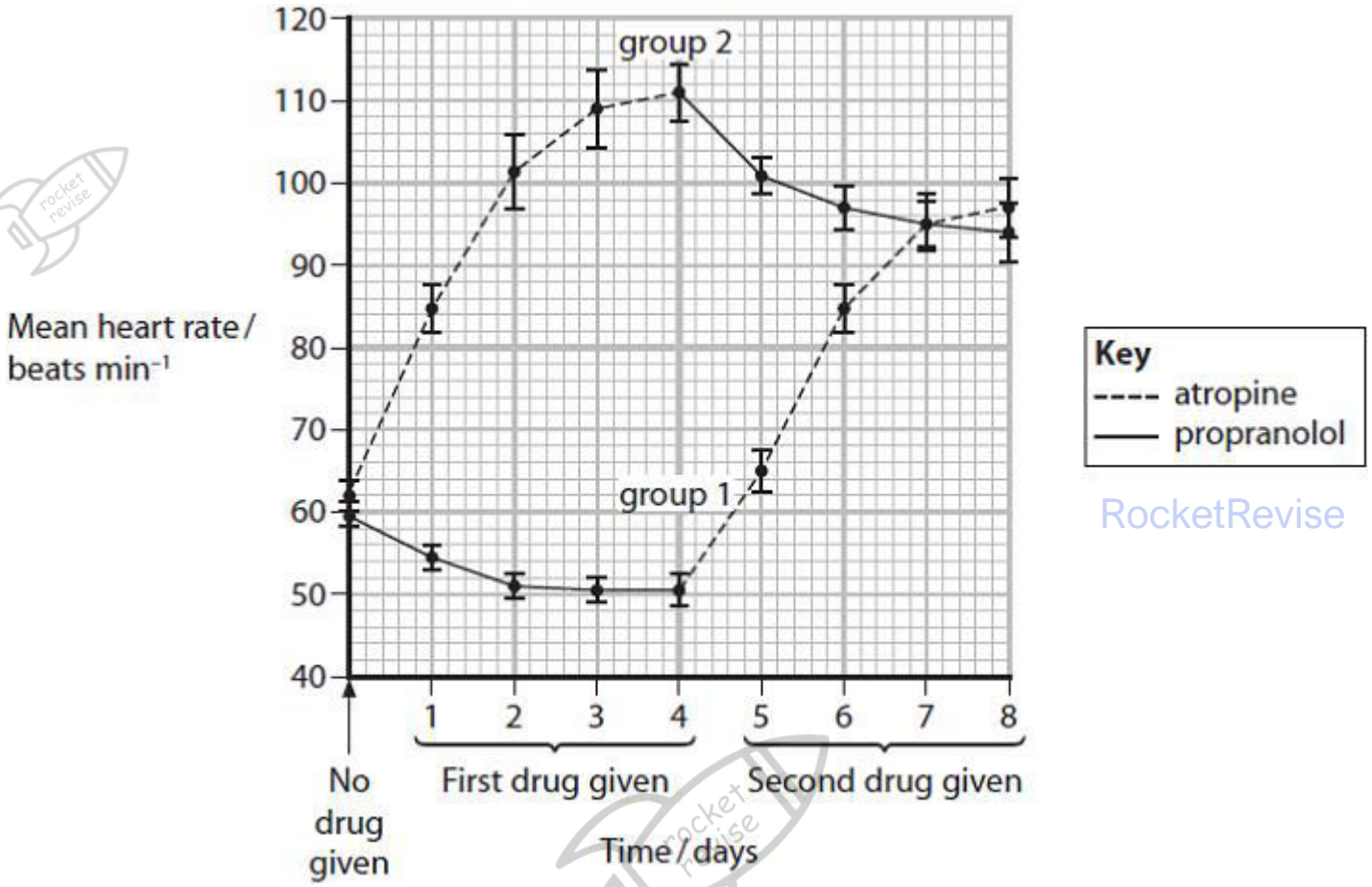
(b) The effect of atropine and propranolol on heart rate was investigated.

One group of people, group 1, were given propranolol each day for four days followed by atropine for four days.
A second group of people, group 2, were given atropine each day for four days followed by propranolol for four days.
The graph shows the results of this investigation.

RocketRevise

www.RocketRevise.com

www.RocketRevise.com



RocketRevise

(i) Calculate the change in mean heart rate for group 2 after 8 days.

(1)

Answer ..... beats min^{-1}

(ii) Calculate the mean length of the cardiac cycle for group 2 on day 4.

(1)

RocketRevise

Answer ..... seconds

(iii) Comment on the conclusions that can be drawn from this investigation.

(3)

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

(Total for question = 9 marks)

RocketRevise

Q16.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Many animals have a heart and circulation.

(a) Molecules such as glucose move by mass transport to overcome the limitations of diffusion in meeting the requirements of an animal.

(i) State what is meant by the term **mass transport**.

(1)

.....

.....

.....

(ii) All the cells in an animal require glucose.
Explain the limitations of diffusion in providing cells with glucose.

(2)

.....

.....

RocketRevise

.....

.....

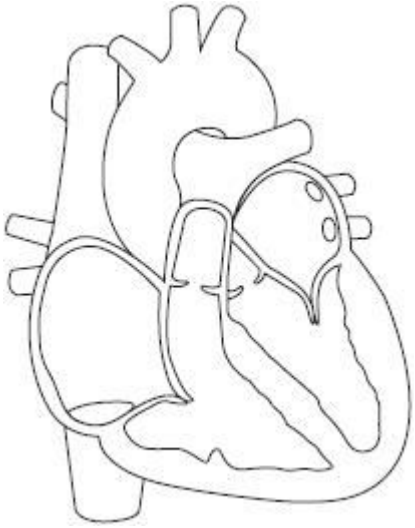
.....

.....

(b) The diagram shows a mammalian heart.

www.RocketRevise.com

www.RocketRevise.com



RocketRevise

Draw arrows on the diagram to show the flow of deoxygenated blood (oxygen-depleted blood) into, through and out of the heart.

(2)

(c) The tunica media is the middle layer in the wall of an artery.

The table shows the percentages of some components in the tunica media of arteries of different sizes.

Component	Percentages of some components in tunica media (%)		
	Large artery	Medium-sized artery	Small artery
Smooth muscle cells	33.5%		60.5%
Collagen	37.0%		12.0%
Elastin	24.5%		1.5%

Explain why the percentages of the components in these arteries are different.

(4)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

RocketRevise

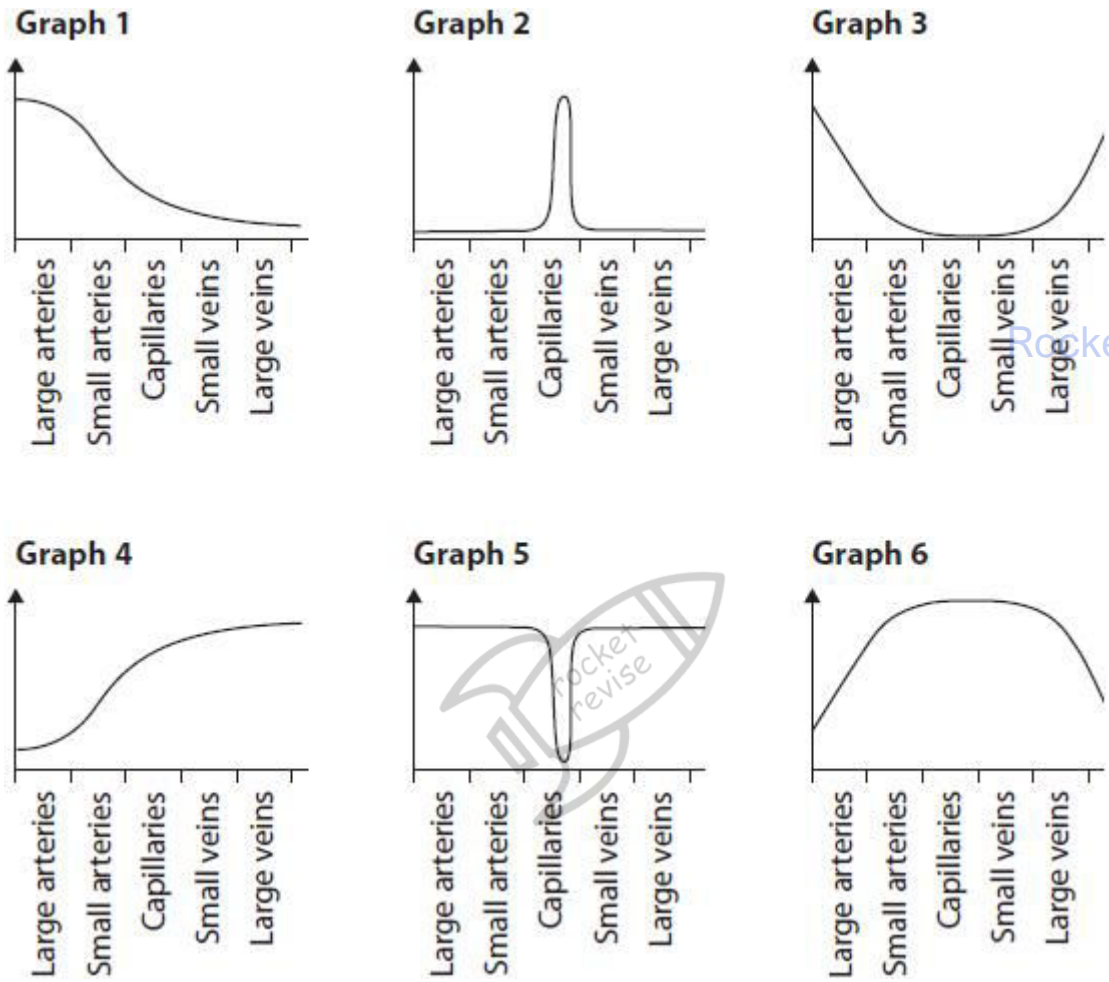
www.RocketRevise.com

www.RocketRevise.com

.....

.....

(d) The graphs show differences in some features of blood vessels.



(i) Which graph shows the diameter of the blood vessels?

(1)

- ☒ **A** graph 1
- ☒ **B** graph 3
- ☒ **C** graph 4
- ☒ **D** graph 6

(ii) Which graph shows the permeability of the blood vessels?

(1)

- ☒ **A** graph 1
- ☒ **B** graph 2
- ☒ **C** graph 3
- ☒ **D** graph 6

(Total for question = 11 marks)

Q17.

www.RocketRevise.com

www.RocketRevise.com

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The circulatory system and the cardiac cycle ensure that all the cells of the body are supplied with their requirements.

(a) The table shows the diameter of the lumens of an aorta and a capillary, and the thickness of their walls.

Type of blood vessel	Diameter of lumen	Thickness of wall / mm
aorta	2.5 cm	2.0
capillary	0.14 μm	0.001

RocketRevise

(i) How many times greater is the diameter of the lumen of the aorta than the diameter of the lumen of the capillary?

(1)

- ☒ **A** 17 800
- ☒ **B** 17 900
- ☒ **C** 178 000
- ☒ **D** 179 000

(ii) Explain why the wall of the aorta is thicker than the wall of a capillary.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

RocketRevise

(b) The table shows the durations of ventricular diastole and ventricular systole at two different heart rates.

www.RocketRevise.com

www.RocketRevise.com

Heart rate / beats per minute	Duration of ventricular diastole / seconds	Duration of ventricular systole / seconds
75	0.53	0.27
200	0.14	0.16

Determine the relationship between heart rate and the duration of ventricular diastole and ventricular systole.

(2)

.....

RocketRevise

.....

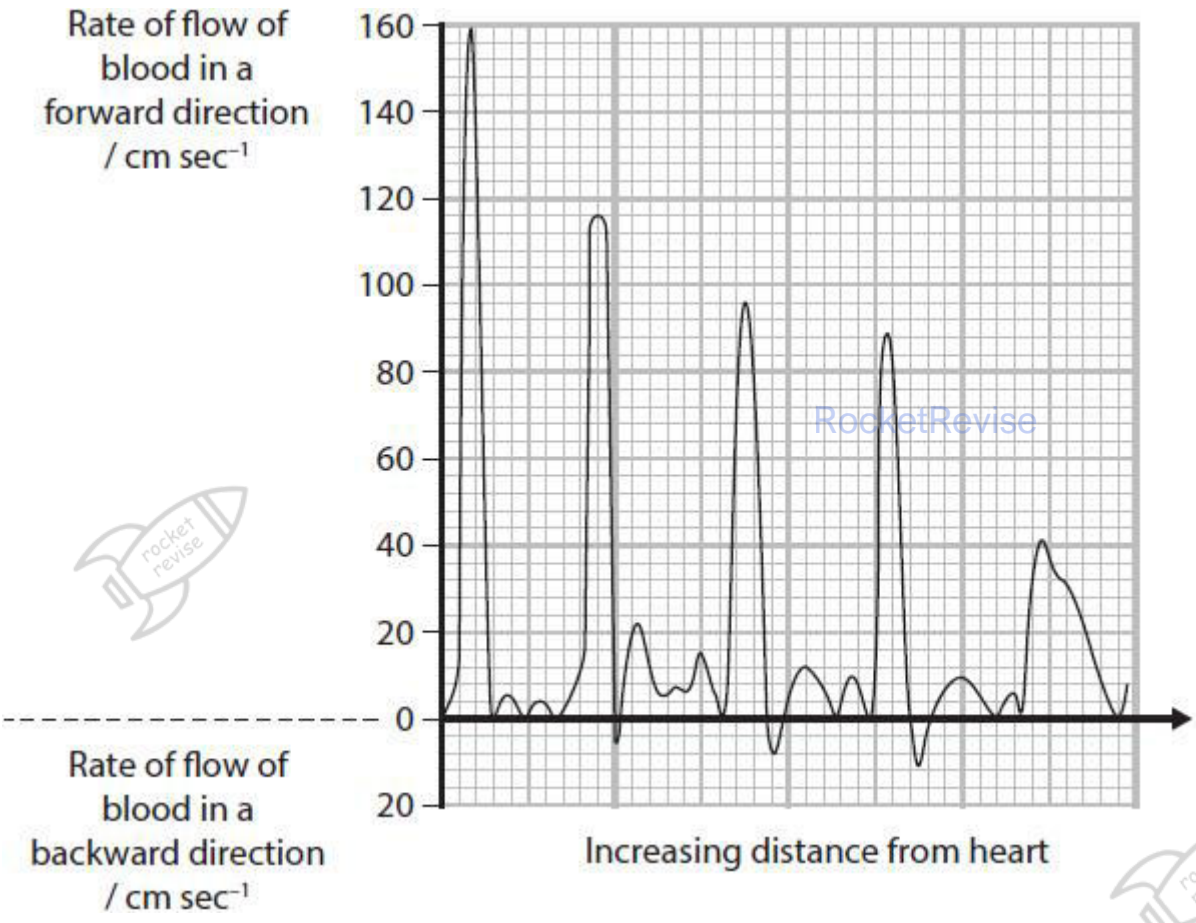
.....

.....

.....

.....

(c) The graph shows the rate of flow of blood through the aorta of a dog with increasing distance from the heart.



www.RocketRevise.com

www.RocketRevise.com

Explain the changes in rate and direction of flow of blood through this aorta.

Use the information in the graph to support your answer.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for question = 9 marks)

Q18.

Red blood cells transport oxygen to the cells of the body.

Read through the following description of how oxygen is transported from the lungs to respiring cells.

Complete the description by writing the most appropriate word or words on the dotted lines.

Oxygen binds to ..... inside red blood cells.

Oxygen binds to this molecule in the lungs where the partial pressure of oxygen is ..... compared with elsewhere in the body.

The oxygenated blood returns to the heart in the blood vessel called the ..... and is pumped to the body by the part of the heart called the ..... .

In the tissues, the ability of this molecule to bind oxygen is affected by the higher concentration of carbon dioxide. This is called the ..... effect.

(Total for question = 5 marks)

Q19.

www.RocketRevise.com

www.RocketRevise.com

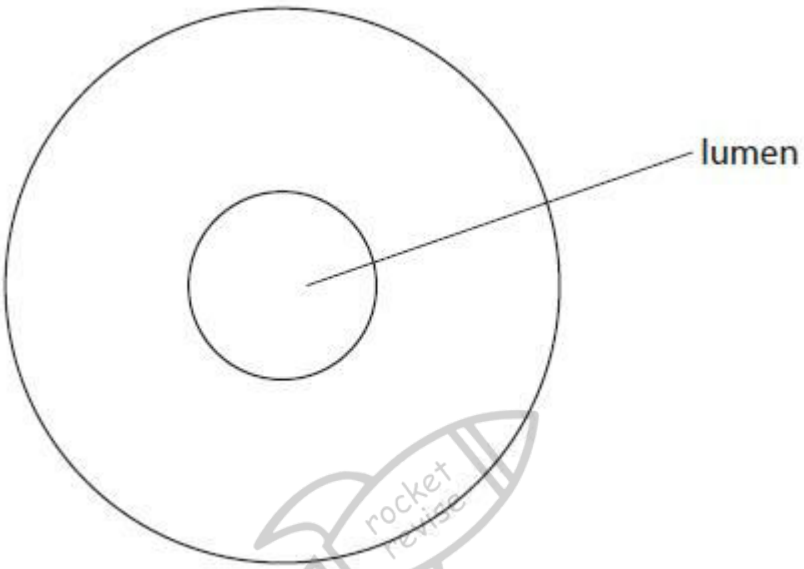
Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The structures of blood vessels relate to their functions.

(a) The diagram shows part of a cross-section of an artery.

Complete the diagram to show details of the structure of an artery.
Label **two** features of an artery on your diagram.

(3)



RocketRevise

(b) The table shows some dimensions of two veins.

Vein	Total diameter of the vein	Diameter of the lumen / mm	Thickness of wall / μm
Vena cava	31 mm	30	500
Small vein	46 μm		4

(i) Which is the ratio of the diameter of the lumen to the thickness of wall in the vena cava?

(1)

- ☒ **A** 0.06 : 1.00
- ☒ **B** 1.00 : 0.02
- ☒ **C** 1.00 : 16.67
- ☒ **D** 60.00 : 1.00

RocketRevise

(ii) Calculate the cross-sectional area of the **lumen** of this small vein.
Use the formula:

$$\text{cross-sectional area} = \pi r^2$$

Give your answer as a whole number.

(2)

www.RocketRevise.com

www.RocketRevise.com

Answer ..... μm^2

(Total for question = 6 marks)

Q20.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Bears hibernate during the winter months, so are inactive for long periods of time.

The photograph shows two bears hibernating.



(Source: © Minden Pictures / Alamy Stock Photo)

Hibernating bears do not develop deep vein thrombosis (DVT).

Deep vein thrombosis is a blood clot that forms within a deep vein, often in the leg.

In humans, short-term periods of inactivity, such as sitting in an aeroplane, increase the risk of DVT.

This is due to reduced blood flow in the veins, causing damage to the walls of the veins.

(a) (i) Which molecule, released from platelets, results in blood clotting?

(1)

- ☒ **A** fibrinogen
- ☒ **B** prothrombin
- ☒ **C** thrombin
- ☒ **D** thromboplastin

(ii) Which pair of molecules are enzymes in their active form?

www.RocketRevise.com

www.RocketRevise.com

(1)

- ☒ **A** fibrinogen and prothrombin
- ☒ **B** prothrombin and thrombin
- ☒ **C** thrombin and thromboplastin
- ☒ **D** thromboplastin and prothrombin

(iii) Which molecule found in the blood is insoluble?

(1)

- ☒ **A** fibrin
- ☒ **B** fibrinogen
- ☒ **C** prothrombin
- ☒ **D** thromboplastin

(b) Pulmonary embolism can be caused by a DVT when the blood clot travels through the circulatory system and blocks the pulmonary artery.

RocketRevise

Symptoms of pulmonary embolism include difficulty in breathing and a rapid heart rate. Suggest why pulmonary embolism can result in these symptoms.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

(c) A study of bears in Sweden discovered that these bears produced less HSP47 protein when hibernating in the winter.

This protein is found on platelets and is involved in the binding of platelets to damaged blood vessels. Discuss whether this discovery could be useful to humans.

RocketRevise

(3)

.....

.....

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

(Total for question = 9 marks)

Q21.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

One risk factor for cardiovascular disease is a short blood clotting time.

One test for measuring blood clotting time is the prothrombin time (PT).

(a) Read through the following passage about prothrombin and the blood clotting process.

Complete the passage by writing the most appropriate word or words on the dotted lines.

(3)

Prothrombin is transported in the .....of a human.
It is converted into thrombin by an enzyme called .....
The substrate for thrombin is .....

(b) Which pair of drugs is used to treat patients who have short blood clotting times?

(1)

- ☒ **A** anticoagulants and antihypertensives
- ☒ **B** anticoagulants and platelet inhibitors
- ☒ **C** antihypertensives and statins
- ☒ **D** platelet inhibitors and statins

(c) The range for a healthy PT is between 10 and 13 seconds.

A person with a PT of 8.5 seconds was treated with a drug to increase the PT time to 10 seconds.
Calculate the percentage increase in PT from 8.5 seconds to 10 seconds.

(1)

Answer ..... %

(Total for question = 5 marks)

Q22.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Haemoglobin transports oxygen and carbon dioxide around the body.

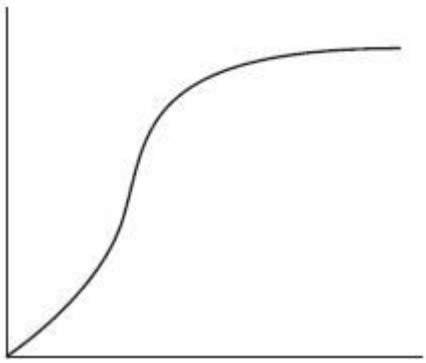
(a) Haemoglobin binds to oxygen in the lungs and releases it in the tissues.

The percentage saturation of haemoglobin plotted against partial pressure of oxygen is called the

www.RocketRevise.com

www.RocketRevise.com

dissociation curve.
The sketch shows the shape of a dissociation curve.



(i) Which describes this dissociation curve?

(1)

RocketRevise

- ☒ **A** linear curve showing a positive correlation
- ☒ **B** linear curve showing a negative correlation
- ☒ **C** sigmoid curve showing a negative correlation
- ☒ **D** sigmoid curve showing a positive correlation

(ii) Which name is given to the effect of increasing carbon dioxide concentration on this dissociation curve?

(1)

- ☒ **A** Benedict
- ☒ **B** Bohr
- ☒ **C** Fick
- ☒ **D** Stahl

(b) Explain how the structure of haemoglobin is related to its function in transporting oxygen and carbon dioxide.

(3)

.....

.....

.....

.....

.....

RocketRevise

.....

.....

.....

.....

(Total for question = 5 marks)

www.RocketRevise.com